



Final Design Presentation For Wearable Glove for American Sign Language (ASL) Gesture Recognition

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Initial Problem & Motivation

- Sign language use is a key marker for those enculturated into a deaf community [1]
- Socioculturally distinguishes being deaf from being hard of hearing [1]
- According to a NHIS survey, **2.80%** of Americans use ASL [1]
 - The same survey **16.12%** of Americans report little to immense trouble with hearing
- The Wearable Glove for ASL Gesture Recognition serves as an effective, low-cost solution to this marginalized group of ASL users



Objectives & Requirements

- Design a portable prototype to function in real-time whilst identifying ASL gestures
- Integrate sensors to interact with an embedded system
- The glove should translate the ASL alphabet and numbers one through ten
- Require that the system predict gestures with $> 90\%$ accuracy
- Comply with ethical safety factors
 - Battery safety
- Make predictions within 65ms
- Maintain low cost

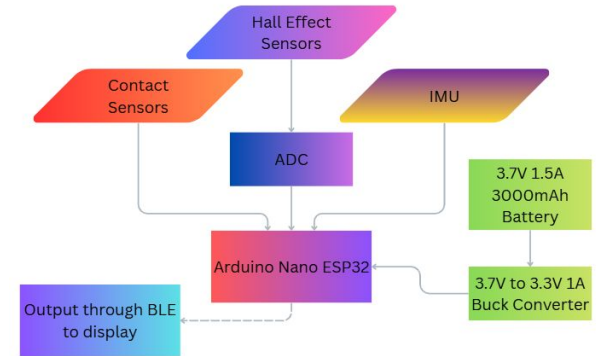


System Overview

- Novel use of sensors pose as a benefit to many different kinds of systems
- “Wearable Glove for American Sign Language (ASL) Gesture Recognition,” works as a solution to translating ASL for a larger audience
- Three different sensory subsystems implemented with a microcontroller using a lightweight model
- The microcontroller displays model predictions via BLE using an application

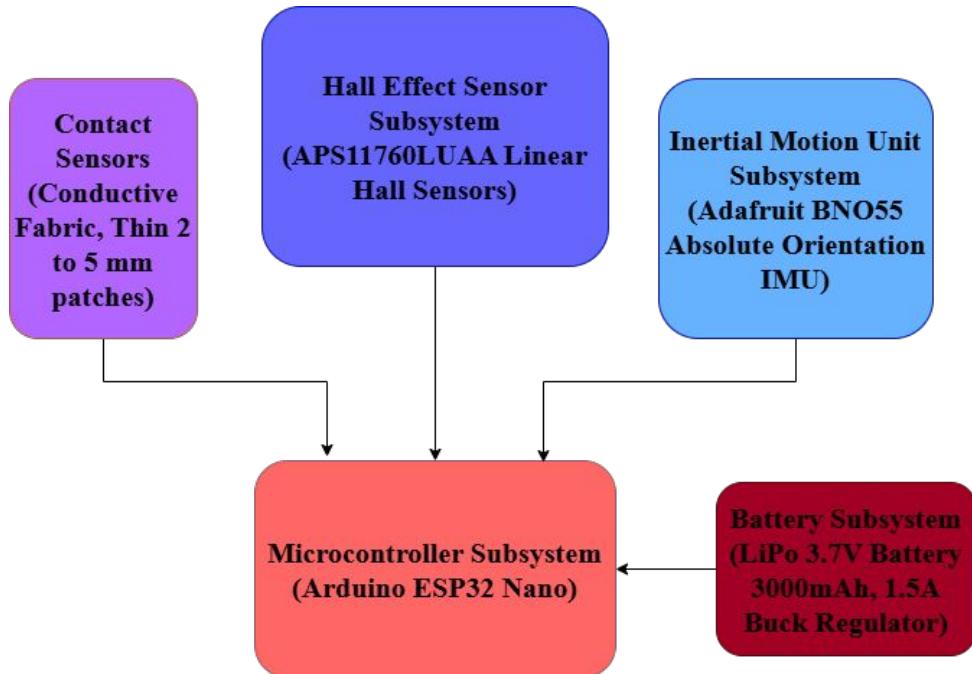
System Overview (Sensors)

- Three different sensor subsystems to determine
 - Bend of the finger
 - Position of the wrist
 - Contact between finger tips
- Sensory data is acquired and handled by a microcontroller
- Performs real-time analysis of sensory data using a lightweight model
- A program interacts via BLE capabilities of the microcontroller



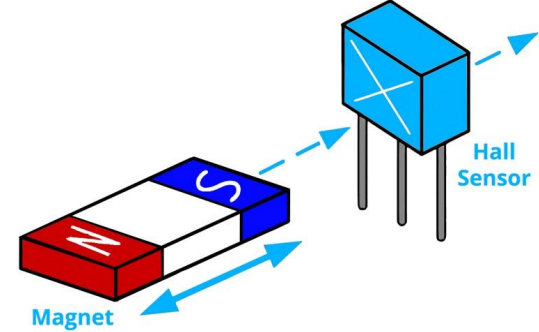
Overview of Hardware Subsystems

- Contact Sensor Subsystem
- Hall Effect Sensors (Finger Bend) Subsystem
- Inertial Motion Unit Subsystem
- Microcontroller Subsystem
- Battery/Power Subsystem



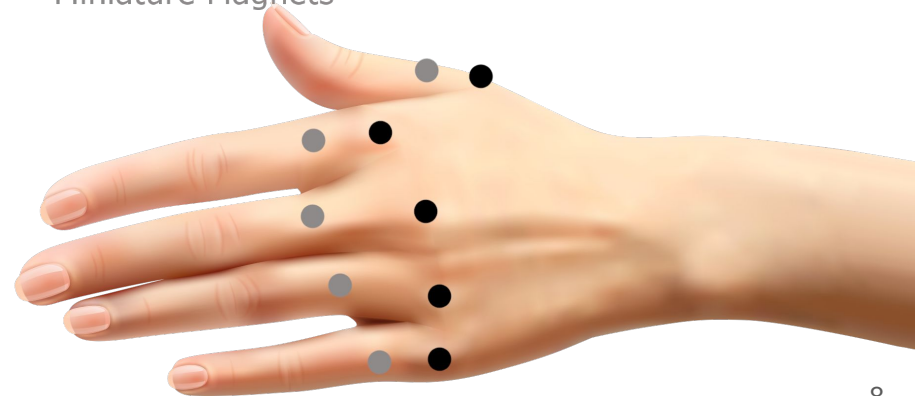
Hardware Design (Finger Bend)

- Hall effect sensors measure magnetic field at a specific point in time
- Hall effect sensors are used to record how far each finger bends from magnet placed along the phalanx
- The sensors are strong enough to notice magnetic force when fingers are stretched apart



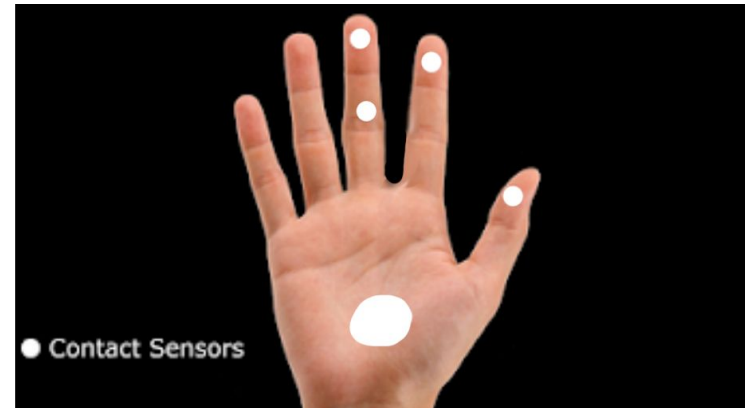
● Hall Effect Sensor

● Miniature Magnets



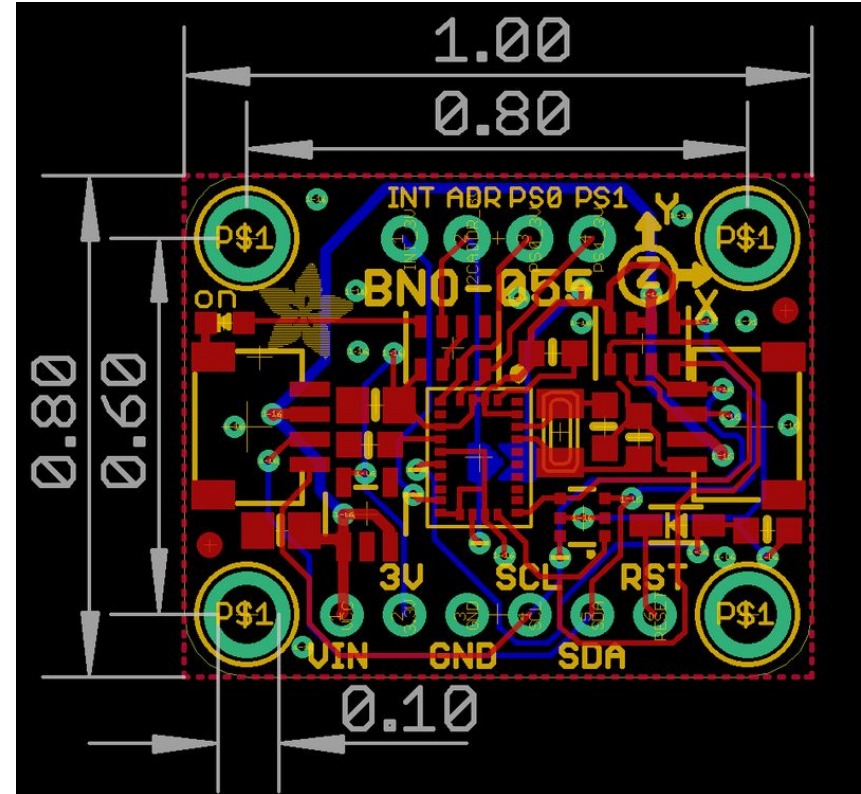
Hardware Design (Contact Points)

- The glove requires that certain finger contacts to increase accuracy
- Sensors are placed along the:
 - Index tip
 - Middle finger tip
 - Thumb (3.3V)
 - Index Rear (3.3V)
 - Palm
 - Under the middle finger
- Examples of gestures affected: 'E', 'F', 'N', 'O', 'R',



Hardware Design (Movement)

- Nine Degrees of Freedom
 - Accelerometer detects acceleration for identifying dynamic gestures
 - Gyroscope measures orientation
 - For example, distinguishes similar gestures such as 'V' and '2'
 - Magnetometer measures speed of the glove to aid in dynamic gesture recognition
- Sends data through SDA





Power Note

- A 3.7V LiPo (3000 mAh) battery with a buck converter is used to directly power the ESP32
- Another option is powering with 5V via USB-C
- 4 - 6 hours of estimated battery life



Evaluation of Alternatives

- Hardware alternatives considered in Top-Level System Architecture
- Considered cost, power, speed, wearability, and manufacturability
- Safety/Welfare weighed higher to ensure safety of the user
- Cultural factors weighed high to ensure to offense is posed to ASL users



Top-Level System Architecture Selection

Weighted Scores

Design 1 — BLE Portable

4.05

Design 2 — Wired

3.55

Design 3 — Website

3.50

Design 4 — Built-in Display

3.95



Top-Level System Architecture (cont.)

- The weighted evaluation above shows the Wearable and Portable Glove with BLE Applications maximize:
 - Safety of the user
 - Clear predictions of gestures
 - Support for end-of-life disposal
 - Minimizes cost
 - Supports ASL communities
- Note: BLE applications can easily be reworked into an app that can translate into alternative languages while a display limits the translation capabilities.
- Design four has the closest score to the first design and offers a competent alternative



Subsystem Architecture Selection

- Weighted decision matrices used to determine best component for each subsystem
- Optimized on different parameters for each matrix
 - Microcontroller: Cost, Complexity, Compactness
 - Finger-bend Sensor: Accuracy, Durability
 - IMU: Compatibility, Redundance
 - Contact Pads: Compatibility, Accessibility

Microcontroller Scores

Arduino Nano ESP32

4.10

ESP32 C5 DevKitC1

3.35

ESP32 PICO V3

3.95

Finger-Bend Sensors Scores

APS11760LUAA Vertical Hall Effect Sensors

4.45

APS11700 Vertical and Planar Hall Sensors

4.35

Bend Labs 1-Axis Flexible Flex Sensor

3.00

Inertial Motion Unit Scores

Adafruit BNO055 Absolute Orientation IMU

4.70

LSM6DS33 3D Accelerometer and Gyro
Carrier

3.80

Contact Scores

Contact Fabric

4.15

Capacitive Glove

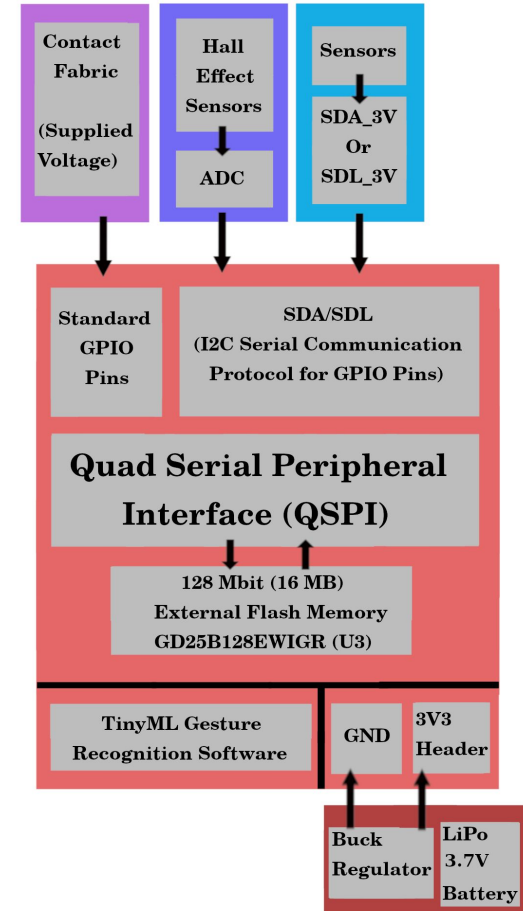
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Aluminum Foil

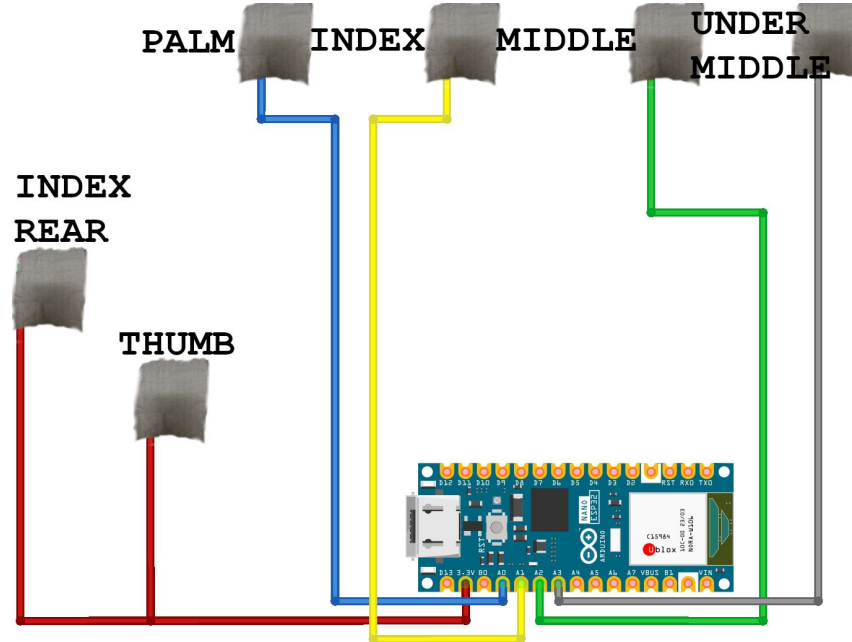
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Hardware Block Diagram

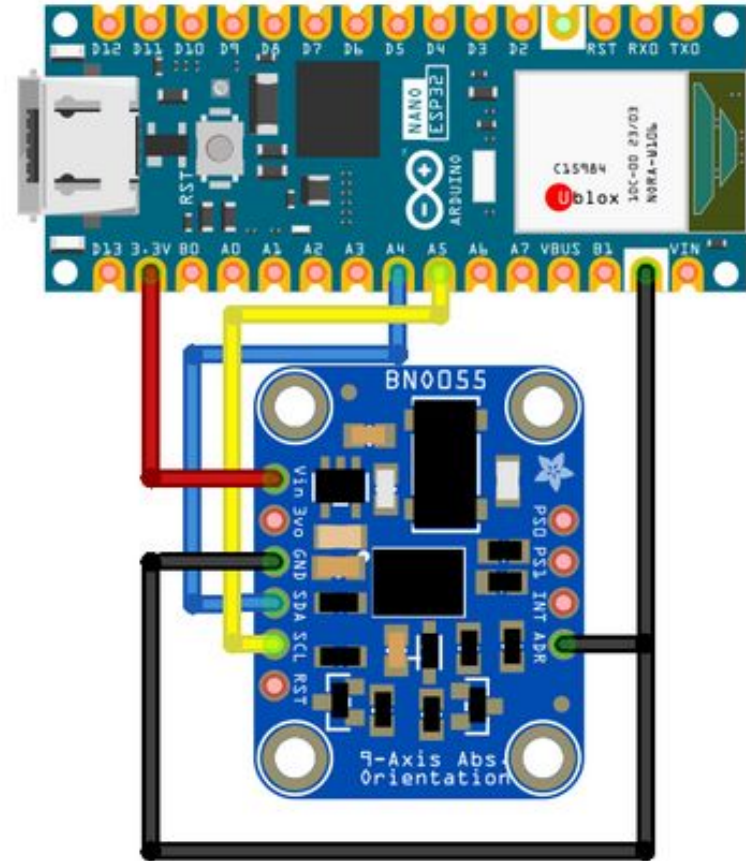
- The hall effect sensors interact with the ADC within their own subsystem
- The light blue represents the IMU subsystem, which interacts with its built in SDA/SDL protocols
- The flash memory on the microcontroller must interact with the Quad Serial Peripheral (QSPI)
- TinyML Gesture Recognition Software will be contained within the external flash memory



Contact Points



Inertial Motion Unit



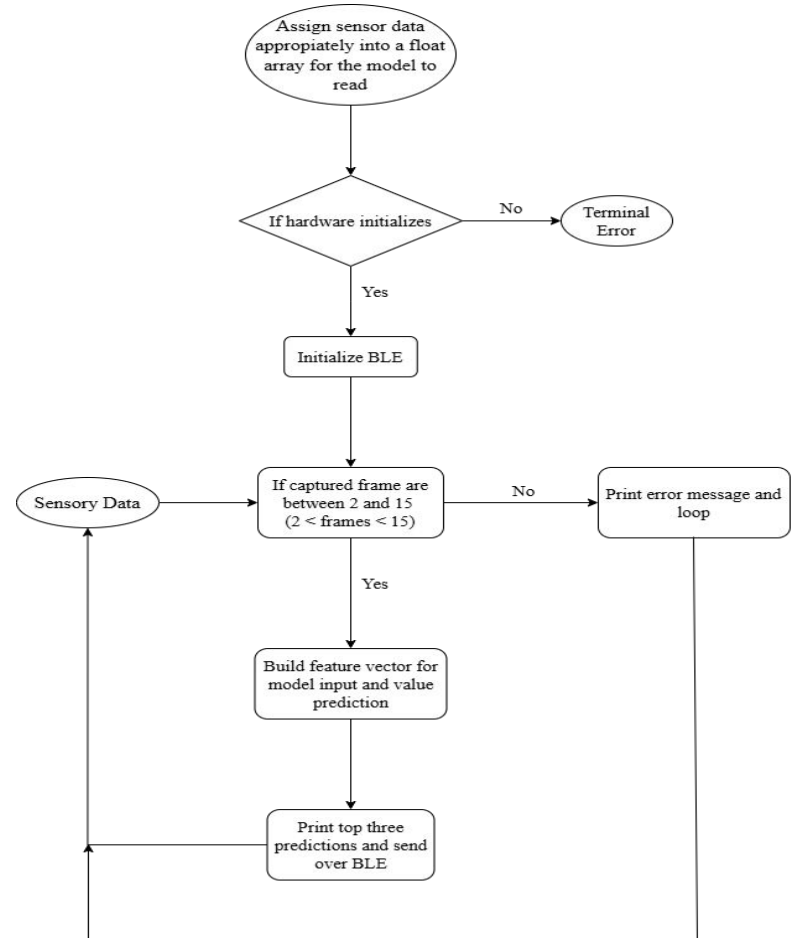


Bill of Materials (High-Level)

Component	Manufacturer	Part Number	Qty	Unit Price (\$)	Total Price (\$)
Microcontroller	Arduino	ABX00092 (Nano ESP32)	1	18.30	18.30
ADC IC (Breakout)	Qoroos	ADS1115	2	14.95	29.90
Hall Sensor	Allegro Microsystems	APS11760LUAA-0PL	5	0.97	4.85
IMU	Adafruit	BNO055	1	34.95	34.95
LiPo Battery	MakerFocus	3.7V 3000mAh Protected	1	15.00	15.00
3.3V Buck Converter	Polulu	D24V10F3	1	8.00	8.00
Conductive Fabric	Adafruit	20cmx20cm	1	14.95	14.95
Neodymium Magnet	Generic	3×8 mm Disc	5	0.13	0.65
Cotton Glove	Generic	Medium Cotton Glove	1	5.00	5.00
Wrist Strap	Generic	Nylon Webbing	1	5.32	5.32
TPU Filament	Generic	Any	N/A	0.10	0.10
				Total	136.99

Software Flowchart

- Pre-Setup:
 - Assign sensor data into array
- Setup:
 - Initialize Hardware
 - Initialize BLE
- Loop:
 - Take in data and check for appropriate frame count
 - Build feature vector and have model predict
 - Print predictions and send over BLE





System Integration

- Microcontroller programmed according to software flowchart
- Seperate program written for real-time data acquisition
- Model is trained using data written to a “.csv” file
- After conversion, model deployed to embedded system with BLE intialization
- Wireless communication application receives prediction



Deployed Model

- Pre-Processing
 - Shuffled data
- Trained a random forest model and decision tree model
 - 80 - 20 train test split
- Displayed metrics for both models
- Roughly 50k lines in data and 2,700 gestures signed



Ethics

- Public health
 - The prototype poses low risk to self, low heat
- Safety and Welfare
 - Insulated, nylon battery pocket protects user
 - Battery life testing
- Global Factors
 - Currently only affects the US
 - Scalable to other countries
- Cultural and Social Factors
 - Benefits ASL using countries
 - Multi-language support for non-ASL users
- Environmental Factors
 - Low risk with proper disposal
- Economic Factors
 - Low-cost materials
 - Cost-effective (~\$150)

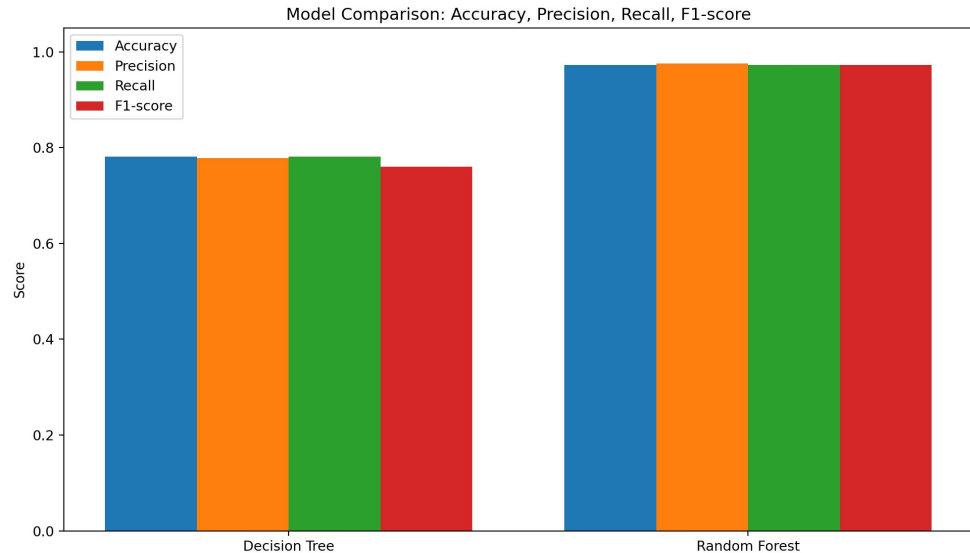


Testing Methodology

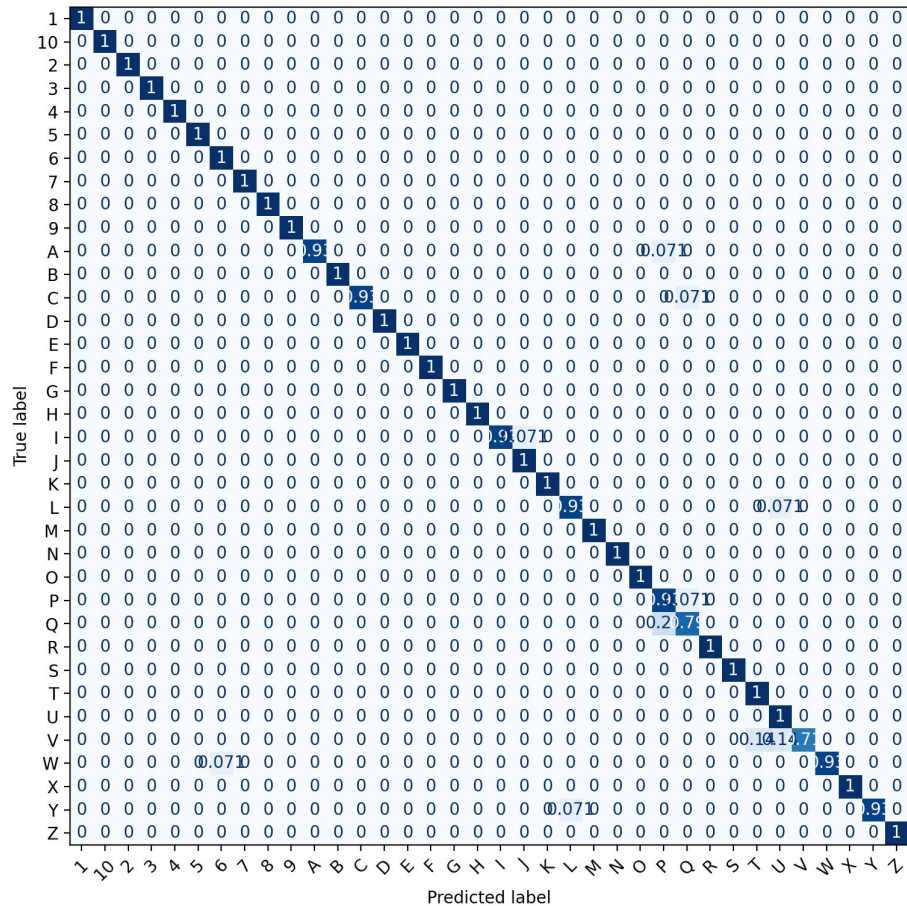
- Accuracy Testing:
 - Each of the 36 gestures were performed 10 times
 - Stability of predictions
 - Confidence of the model
 - Leniency on prediction times
- Battery life and user safety:
 - Battery life measured with stopwatch
- Time to make prediction:
 - $\text{Response Time} = (\text{Time of Output}) - (\text{End of Reading Time})$

Experimental Results and Analysis cont.

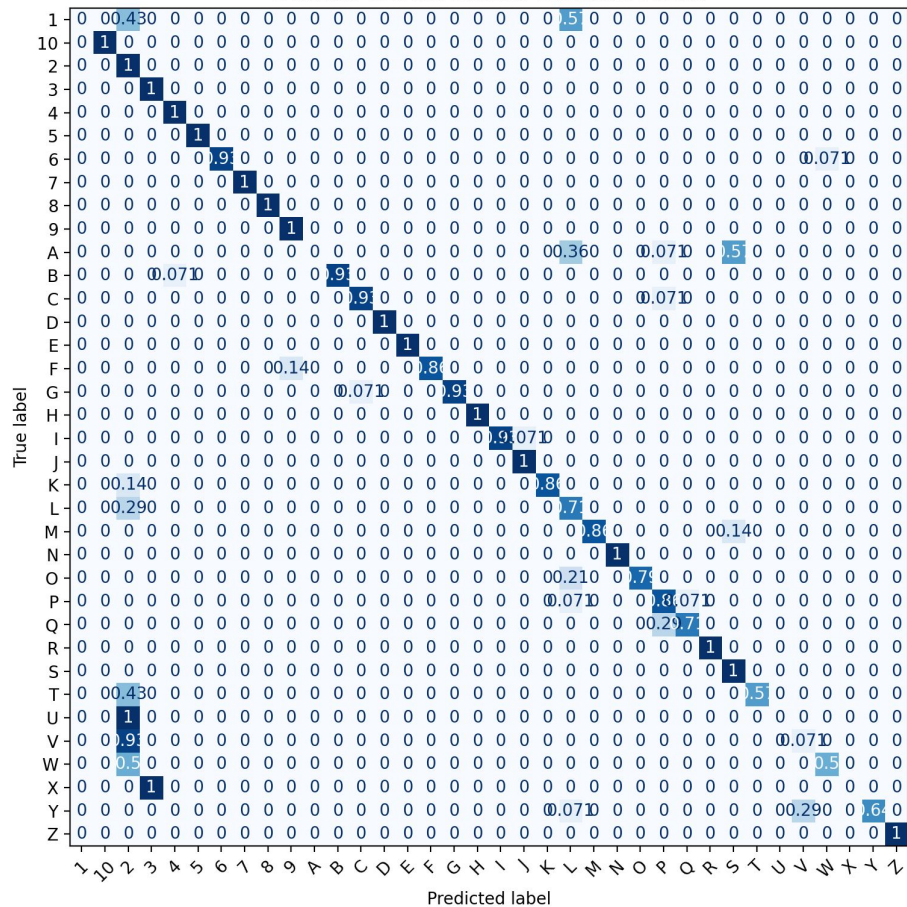
- After model training we graphed the results of the decision tree model and random forest along with their respective F-1 scores, recall, accuracy and precision
- Random Forest provided better results
- Decision Tree performed adequately, but did not meet design specifications



Random Forest Normalized Confusion Matrix

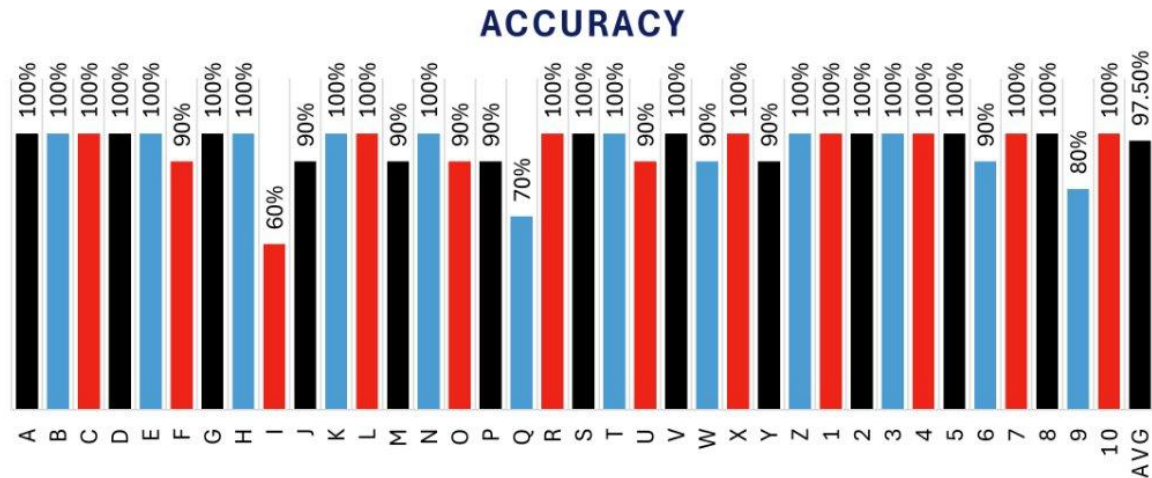


Decision Tree Normalized Confusion Matrix



Experimental Results and Analysis cont.

- For decision tree, performance too varied and did not meet system requirements
- Random forest model performs strong with similar gestures
 - ‘Q’, ‘I’, and ‘Y’ performed weaker than others, but still maintained high performance metrics





Broader Impacts of the Project

- Safety/Welfare and Cultural factors are expected to have the most positive influences:
 - Extreme caution was maintained ensuring accuracy for cultural factors
 - Readily available public sources were used to verify
 - Pocket prevents heat and ensures no direct contact between battery and skin.
 - Safe charging and battery chassis necessary for commercial units
- Other factors weren't as important, but can still expect positive influences:
 - The glove is cost effective to produce
 - \$150 to produce each unit
 - The glove uses minimal electronics that will have a minimal environmental impact if disposed of properly

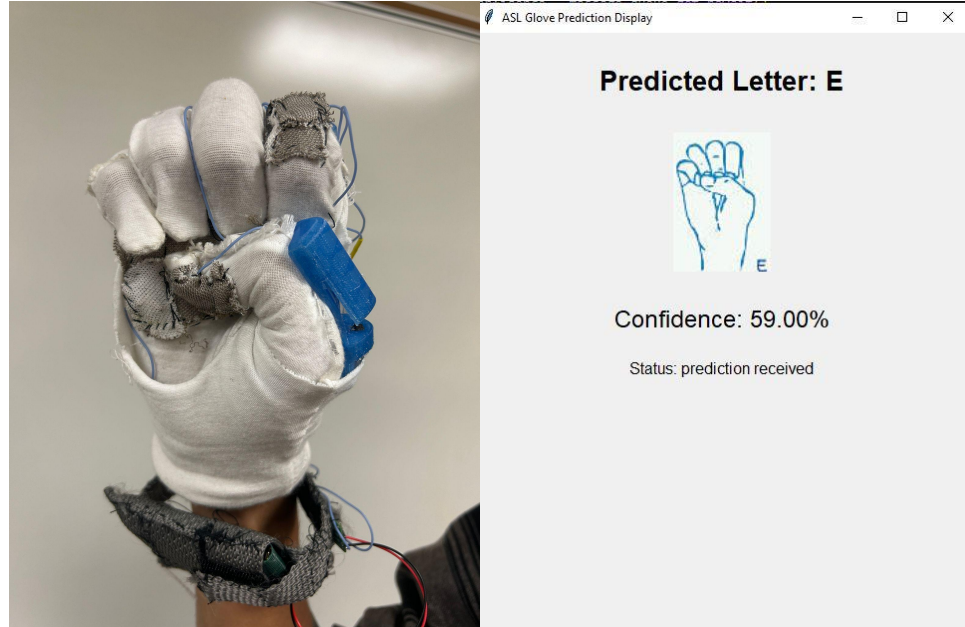


Future Work

- Full word gesture recognition
 - Duplicate the glove to the left hand and estimate relative spatial position to the users' body
 - Requires replacement of the microcontroller to allow for even connection on both hands (Perhaps back or chest)
- Multi-Language support
 - Train on different data and change the glove slightly
 - Some sign languages may not use the index as much as american sign language
- Further work can be achieved with further testing and time

Conclusion

- Developed real-time prototype
- Achieves an accuracy of 97.50% and maintains health and safety with adequate battery life
- The glove was designed with long product life in mind
- Compact and efficient





Acknowledgements

- Thank you Dr. Rizk for assisting us with the design process and to enhance our reports and presentations.
- Thank you UTTyler for providing lab space for the creation of our prototype and the testing of the prototype



References

[1] Mitchell, Ross & Young, Travas. (2022). How Many People Use Sign Language? A National Health Survey-Based Estimate.

Q&A